



PIKAR: A Pixel-Level Image Kansei Analysis and Recognition System Based on Deep Learning for User-Centered Product Design

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Abstract. Can machines learn to perceive products like humans? Kansei Engineering has been developed to connect product design and human perception. While conventional Kansei Engineering Systems exhibit a high dependency on manual extraction of design elements and thereby are restricted to validity issues, we present a Pixel-level Image Kansei Analysis and Recognition (PIKAR) system that applies deep learning to extract and analyze the formation of human perception towards product designs automatically. Method validation is performed based on the evaluation of cosmetic packaging's kawaii. Two neural nets trained on 1,414 images, labeled by eight participants based on their perception of kawaii (1–5 Likert Scale), have achieved a better prediction than test persons. The implemented neuron analysis methodology for Kansei analysis points towards consistency with previous experimental kawaii studies and gives insight to individual differences. This work addresses the possibility of applying deep learning to support product design and user experience researches.

Keywords: Deep learning · Consumer product design · User experience · Kawaii

1 Introduction

The word “Kansei” originates in Japan and refers to the customers’ feelings and perception towards a product. Kansei engineering has been developed as the translation technology between product design parameters and human perception and has maintained a significant role in the consumer-oriented product development paradigm [1]. For decades, the construction of Kansei Engineering System (KES) has relied on the manual definition and extraction of significant product attributes, i.e. the design elements [2–7]. Two typical types of element-level KES exist in this domain: a backward KES focuses on predicting Kansei from products and requires manual extraction of design elements prior to mathematical modelling; and a forward KES performs systematic analysis of design elements and identifies factors affecting Kansei. The classic KES can be summarized as an inductive system based on small samples and manual extraction, but it has the following problems to be solved [8]. Firstly, a typical KES usually only uses no more than 100 images of the product as a sample. Insufficient sample size and

sample selection bias will lead to doubts about the validity of the conclusion. Secondly, the process of selecting key design elements often relies on expert advice, but human cognition has inevitable limitations. In addition, the selection of key features is often vague, so it is difficult to guarantee the comprehensiveness of the conclusions. Thirdly, the current system can only process manually processed images, not the original images. This means that the classic KES is difficult to apply on a large scale in the commercial field.

To circumvent the aforementioned constraints, we propose PIKAR system, a Pixel-level Image Kansei Analysis and Recognition system based on deep learning. Contrary to element-level KES, our method is an end-to-end, pixel-level approach that recognizes product Kansei from raw pixel distribution without prior feature handcrafting or extraction. Besides, we propose a deep learning based Kansei analysis method: the neuron analysis methodology incorporating net visualization techniques to identify features affecting Kansei for future design improvement. The method validation is performed based on cosmetic packaging's kawaii evaluation. In addition, autonomous extraction of product Kansei is crucial to customer-oriented product design and precision marketing. First, 1414 images of cosmetics labeled by 8 test persons according to their perception of kawaii (1–5 Likert scale) are used for training the deep neural nets KawaiiNet and KawaiiScoreNet. Results show that KawaiiNet achieves an accuracy of 84% in identifying kawaii design of cosmetic packaging; and KawaiiScoreNet predicts the kawaii values with an accuracy of 90.6% within a 1-point tolerance and 69.3% within a 0.5-point tolerance. The latter demonstrates a better prediction accuracy than the single test person (90.6% vs 56.1%, 69.3% vs. 28.3%). Second, we explore the deep representatives learned from images to identify design features affecting the perception of kawaii. Our results correspond with previous kawaii studies [9, 10] depicting the principal five factors causing kawaii feelings and give insight to subconsciously perceived product features and factor interaction.

There are mainly three contributions of this work. Firstly, for the first time, we propose a new method to extract subjective attribute “Kansei” from pixel-level images automatically. The system no longer uses design elements as the smallest input unit of the inductive engineering system, but directly uses image input to facilitate the direct use of big data. Secondly, we establish a neural net for predicting kawaii value of cosmetic packaging and a neuron analysis method for detecting features affecting Kansei within the deep learning framework, which is more insightful than the traditional method, and can also be applied to the study of individual differences. Thirdly, this work sheds light on a data-driven research paradigm in the field of product design and human factors, providing new perspectives to the current methodologies and theories.

2 Related Work

Extensive studies on product Kansei prediction are based on basic design elements such as the color combination [4], the texture pattern [2] or shape of a certain part [5]. Kansei Engineering System has been proved useful in industrial design [6, 7], packaging design [3, 4], and even interior design [11]. In comparison, less work has been focused on modelling product Kansei directly from raw images. Gong and Rau have proposed a system

structure in this domain, but the system lacks further validation [8]. Another relevant study is DeepSentibank that propose a deep learning based, large-scale visual sentiment ontology using Adjective-Noun Pairs [12]. This model uses one million images to train convolutional neural nets and learns to recognize objects with their perceptual attributes. But their focus is rather on general items (e.g. flowers, faces) than on a specific product. Other studies perceive abstract attributes of images, such as photographic aesthetics [13], popularity of images [14], memorability [15] and artistic styles [16] and have provided strong evidence that deep learning is the most adaptive and suitable approach for recognizing vague image attributes.

In terms of Kansei analysis, the purpose is to discover the visual cues of a product that cause the formation of certain type of Kansei. Several research in the field of net visualization have revealed the connection between neuron activations and the appearance of certain visual cues. Girshick et al. point out that each neuron in a deep neural net captures one or two certain visual patterns [17]. Neurons on the lower-level layers (e.g. the first one or two hidden layers) represent fundamental visual properties such as color and edges, while high-level neurons fire to a specific shape, pattern or a category of object. Yosinski et al. demonstrate the “expectation” of each neuron by creating the images that maximize the activation of the certain neuron [18]. This inspires us that neuron activation patterns may be suitable for observing the visual characteristics of a certain image set. The contribution of design features to the formation of Kansei can theoretically be quantified, analyzed, and visualized within the framework of deep learning by interpreting the neuron activations. This system concept is briefly described in [8] and this study performs further implementation and validation of the concept.

3 Recognizing Kansei

This section introduces the construction of a pixel-level Kansei prediction system. In the following experiments, we choose the evaluation of cosmetic packaging’s kawaii as the research object. The considerations for this choice are listed as follows: (1) Prone to compare and validate: Kawaii is a well-researched type of Kansei in comparison to other alternatives, especially among Japanese researchers (2) High practical value: Kawaii packaging of cosmetics has a significant effect on the purchase decision. Brands like Benefit, PAUL&JOE, Loretta, ETUDE are known for such design strategies. (3) Less noise of data: Cosmetics’ images have favorable characteristics for image analysis, such as standard white background, single object of interest per image and adequate image resolution.

3.1 Method

For all experiments, we use a modified AlexNet [19] as the base model for transfer learning: we start with weights pre-trained on a larger dataset to increase model accuracy. Since we assume the dataset to be small in real practice (~1000 images in this study) and different from the original dataset, namely ImageNet, the following modifications are made to limit model overfitting:

- The classifier is trained from activations in the third pooling layer (pool3) since lower layers contain less dataset-specific features.
- Three fully connected layers with 40 nodes each are inserted before the classifier to obtain the feature vectors.

All images are resized to 256×256 pixels while retaining the original aspect ratio: in non-squared images and a white background is added. We perform random cropping to an image size of 227×227 pixels and random mirroring to augment dataset size. We use SoftmaxLoss as the loss function for kawaii classification (kawaii vs. not kawaii) and EuclideanLoss for regression (value of kawaii). The models are trained with Caffe [20].

3.2 Datasets

All cosmetics images are downloaded from the official websites of 25 international cosmetic companies. A total of 1,414 identical cosmetic images are collected, including products for skin care, body care, makeup for eyes, brows, face and lips, perfume and nail polish. All images have light-colored simple background with one product placed in the center of the image. The minimum resolution we accept is 230×230 pixels which is larger than the minimum input requirement of the network.

Next, 8 young females (Mean age: 24, SD: 1.51) representing the consumer demographics are recruited for the kawaii labelling task. All participants claim to have experiences in purchasing cosmetics from both online and shops. The number of the participants is decided based on a pilot experiment. During the pilot experiment, 15 participants are asked to label five cosmetics images' kawaii value using Likert-5 scale. Results show that the average kawaii values for all 5 test images change drastically when total number of testers is small but later become stable with relatively small differences from the total average scores (<0.5), starting from the 8th testers.

The 8 participants are given 2 h for labeling kawaii values for 1,414 images, allowed with brief breaks after sets of ~ 200 images to avoid fatigue. The labeling test is performed on a MatLab interface with 6 images per page, 5 radio buttons below each image and a next button. The images are displayed with an identical image size. Figure 1 shows the distribution of the final averaged kawaii values and the corresponding standard deviation of the dataset. The most and least kawaii samples are displayed on the right side.

Finally, the total dataset for the classification experiment comprise 500 most kawaii images and 500 least kawaii images while the regression experiment uses all 1,414 images. The dataset is divided into sub-divisions, namely the training set, validation set and test set, with a ratio of 7:1.5:1.5. The division is performed randomly.

3.3 Result of Experiments

Table 1 shows that KawaiiNet appears effective at predicting kawaii cosmetic designs (Overall accuracy: 84% vs. random: 50%). The columns report results on different subsets, showing a higher classification accuracy on images with obvious Kansei than the ambiguous ones (20 smallest SD: 85% vs. 20 largest SD.: 65%).

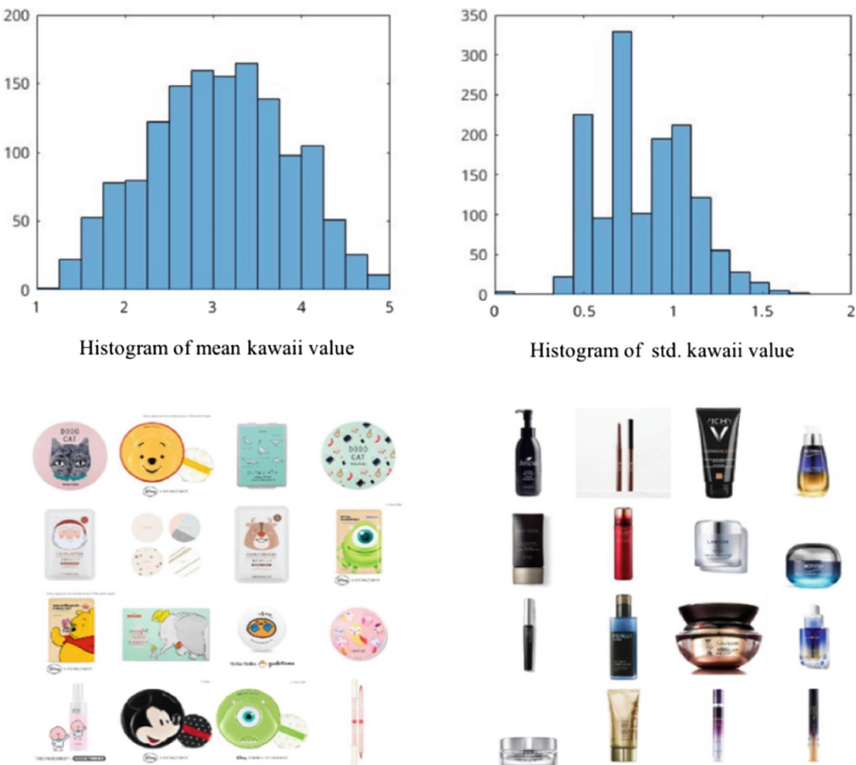


Fig. 1. The distributions of the kawaii values and the standard deviations are shown on the top. The dataset is balanced with kawaii and non-kawaii instances. The final kawaii value for each image is an average score of all participants. The mean kawaii value of all images is 2.9981 (SD = 0.79). Top kawaii samples and non-kawaii samples are shown on the bottom.

Table 1. Accuracy of *KawaiiNet*

	Overall	Top 20	Bottom 20	Large SD 20	Small SD 20	Random
Accuracy	84%	95%	90%	65%	85%	50%

Table 2 shows complete test results of *KawaiiScoreNet*. The predictions are compared with random guess and one human tester on four tolerance scales. Random guess is generated from a normal distribution (mean: 2.9981, SD: 0.79) close to the origin kawaii value distribution of the dataset. The human predictions are obtained from the test person who gives the closest scores to the average kawaii scores of eight participants. Results show that *KawaiiScoreNet* outperforms the single human predictor with a large margin in all scales, achieving a 90.6% confidence in providing prediction within 1-point tolerance and 69.3% within 0.5-point tolerance. Even after integer conversion of the predictions (since the regression results are not in Likert-5 scale in comparison

to human), KawaiiScoreNet still gives kawaii scores that better represents the averaged Kansei than a single test person does.

Table 2. Accuracy of *KawaiiScoreNet*

Tolerance	Random guess	Best human predictor	Kawaii ScoreNet	Kawaii ScoreNet (Integer)
<0.1	6.2%	7.5%	15.1%	9.0%
<0.3	18.8%	23.1%	48.6%	20.6%
<0.5	30.8%	28.3%	69.3%	55.7%
<1	57.8%	56.1%	90.6%	85.3%

Figure 2 compares top 20% most and least kawaii images selected by PIKAR and all human testers, respectively. A preliminary Turing test conducted with a small group of people shows that most people we asked cannot distinguish the “machine-selected kawaii” from the “human-selected kawaii”. Though it is only an early observation, we can see that PIKAR’s ability in predicting kawaii and non-kawaii cosmetics is hardly distinguishable from that of a human.



Fig. 2. The top 20% and bottom 20% images selected by human and by machine from the perspective of kawaii within the test image set ($n = 212$). The left ones are selected by human and the right ones are predicted by machine. Machine’s ability in predicting kawaii cosmetics is hardly distinguishable from that of humans.

4 Analyzing Kansei

This section introduces our second application of deep learning in Kansei engineering. We present a neuron analysis approach to identify features that cause the formation of Kansei and support future product design.

4.1 Method

As recorded in previous studies, one neuron fires on to one or two certain visual modes [17, 18]. This means the total count of activations for a certain neuron indicates the appearance of certain visual cues in a collection of images. If these neurons are identified, which are more activated in image set A and, in comparison, far less activated in set B, they allow to describe the critical visual characteristics that distinguishes set A from set B. This briefly describes the core concept of our methodology. This concept is implemented and validated by the following two experiments. The first aims to detect kawaii factors and the second to understand individual differences in kawaii perception.

As our interest lies in medium-sized regional visual patterns (i.e. neither too general nor too detailed), we focus on neurons from the pool3 layer, the max-pooled output of the third convolutional layer with a receptive field of $99 * 99$ pixels. A total of 13824 neurons are included in the selected layer. For the first experiment, 150 kawaii cosmetics images and 150 non-kawaii images are input to the KawaiiScoreNet separately for neuron activation computing and the total number of each neuron's activations is counted. Neuron activation count differences exceeding 50 between the datasets (top 0.7% large disparities) are selected as the crucial neurons for kawaii perception and further clustering is performed to reduce redundancy. Clustering operation is based on the distance of each neuron's activation vector. The activation vector is a 150-dimension vector composed of one certain neuron's activation values on the kawaii dataset. Second experiment, on the other hand, picks two testers who demonstrated different rating behaviors in the previous labelling task. The opposite image sets consist of 150 kawaii images each, but selected by different testers. We perform the same computation with the datasets as the first experiment except for the crucial neurons selection step. Since the disparities of individuals on perceiving the same type of Kansei is smaller than that of two opposite Kansei types, we adopt a less strict selection policy that captures neurons which are among the top 30% most activated neurons in image set A but fall below average in image set B. This policy results in a total number of 42 and 18 neurons representing the critical visual characters of each tester.

4.2 What Makes Cosmetic Packaging Kawaii?

Figure 3 shows the kawaii traits discovered by neuron analysis method. The graph exhibits the 100 crucial neurons identified to strongly affect the kawaii feelings and the corresponding clustering results. These crucial neurons are divided into 3 categories and 13 sub-categories. Typical images representing each sub-category are displayed below the graph. Each row contains 10 top-scoring images of one random neuron in the sub-category and demonstrates the dominant visual traits for which neuron is activated. Some rows share fewer clear characteristics than the others due to the fact that each neuron

fires to more than one visual cues. This means factor interactions are considered in this method. Naming of the categories are performed by the authors. The resulting three principle factors affecting kawaii are pattern, color and shape. And the 13 sub-categories are named as follows:

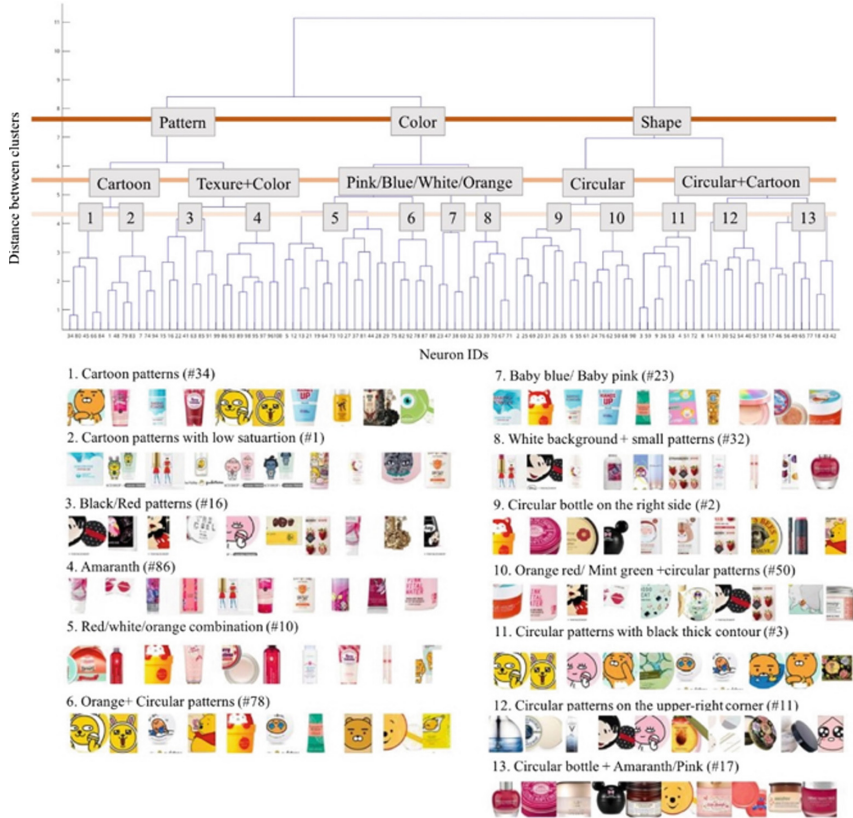


Fig. 3. Top 100 neurons affecting kawaii feelings are detected with neuron analysis method and are clustered into 3 categories and 13 sub-categories. Typical top-scoring regions corresponding to each cluster are displayed. The names of the categories are given according to the dominant traits.

- Cartoon patterns
- Cartoon patterns with low saturation
- Black/red patterns
- Amaranth
- Red/white/orange combination
- Orange + Circular patterns
- Baby blue/baby pink
- White background + Small patterns

- Circular bottle on the right side
- Orange red/mint green + circular patterns
- Circular patterns with black thick contour
- Circular patterns on the upper-right corner
- Circular bottle + Amaranth/pink

Comparing the results to the five principle factors identified to cause kawaii feeling, our method has covered three of them, namely, pattern, color and shape. The remaining two factors that are omitted in this study are size and tactile sensation. The reason for that is all testing images are presented virtually on a computer screen and displayed with the same size. This implies further labelling process improvements.

Other findings in sub-categories concord with the kawaii rules listed in previous studies [9, 10, 21], such as employing warm color scheme, high brightness colors, circular patterns and mammalian cartoon patterns. It is possible that our method cannot provide a comprehensive list of factors, but the dominant ones are detected. Besides, the deep representatives are more powerful in detecting subtle differences and even the subconsciously perceived product features and their factor interactions. See subcategory 8 and 11.



Fig. 4. Neurons reflecting individual difference in kawaii perception between tester A and B.

4.3 Individual Differences on the Perception of Kawaii

Figure 4 shows the most distinguishable visual cues detected with neuron analysis method to represent two selected test persons' kawaii preferences. 42 and 18 neurons are detected respectively, and clustering results show that tester A prioritizes shape while tester B focuses on patterns. More specifically, cosmetics with cylindrical bottles, and ones with wide top and narrow bottom can trigger the kawaii feelings in tester A. Tester B, in comparison, gives higher kawaii scores to the packaging printed with cartoon patterns. The disparities can be clearly observed from the top-scoring images. Additionally, two testers are found to have different reflection models between color and kawaii feelings. Flesh color, red/blue with high saturation and blue-green color are among the kawaii colors for tester A, while bright yellow, mint green and yellowish green are perceived as kawaii by tester B. The same behavior is observed in the labelling experiment. Though a more careful selection of parameters and policies can be used during the neuron selection process, the current results have shown the possibility of applying deep learning in researches such as individual differences in visual perception.

5 Conclusion

Can machines perceive products like humans? Our study is among the first attempts to answer this question using the deep learning technology. In this paper, we explore how low-level pixel distribution of product images can be interpreted to high-level human perception through deep learning. A case of evaluating the kawaii feelings of cosmetic packaging has shown the capability of machine in predicting abstract human perception. We further present how the neural nets can be analyzed to support design and provide insights into individual differences. The results correspond to previous experimental studies and exhibit interesting kawaii traits in the cosmetic packaging design. The proposed technology has possible applications in customized product recommendation systems, design-supporting systems and even in social robotics which mimic human perception.

Yet the answer to the aforementioned question is still open. Several directions remain for future researches. An important question is whether this method is applicable to variations in products and types of Kansei. To what extent can we reuse the neural nets trained with cosmetic packaging? And in what circumstances can we keep the same network structure? We have conducted several preliminary tests with small data size in this direction. The results suggest that a weight-level modification (i.e. keep the model structure and retrain the weights) may be required for cases such as "Luxury perception of cosmetic packaging" or "Kawaii perception of beverage bottles"; but a model-level modification (i.e. change the model structure) tends to be necessary for other applications such as "kawaii value of a girl's face". Neuron analysis for product design improvement is another interesting direction of exploration. Deep learning is well acknowledged as the powerful prediction tool, but this study has explored its ability in extracting visual information for design analysis.

Finally, this study has addressed the potential of applying deep learning in user-centered product design. Our method represents a new approach to gain customer insights with less restriction to the sample size and validity issues and is completed driven by data itself.

Acknowledgement. This work was funded by Tsinghua University Initiative Scientific Research Program 20193080010.

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